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E. Easy Tree Queries

time limit per test: 1 second

memory limit per test: 1024 megabytes

There is a tree with N nodes. The tree is rooted at a given node R .

You will be given Q queries. In each query, you are asked to find the size of the subtree of a given node X .

Input

The first line contains two integers N, R ($1 \leq N \leq 2 \times 10^5, 1 \leq R \leq N$) — the number of nodes and the root of the tree.

The next $N - 1$ lines each contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — representing an bidirectional edge between nodes u_i and v_i .

The next line contains an integer Q ($1 \leq Q \leq 2 \times 10^5$) — the number of queries.

The next Q lines each contain a single integer X ($1 \leq X \leq N$) — the node whose subtree size you need to compute.

Output

For each query, print a single integer — the size of the subtree of node X .

Examples

input	Copy
<pre>4 1 3 1 1 2 4 2 3 1 4 2</pre>	
output	Copy
<pre>4 1 2</pre>	

input	Copy
<pre>5 3 1 2 5 3 3 2 2 4 5 3 5 4 2 1</pre>	
output	Copy
<pre>5 1 1</pre>	

CSE221 Assignment 05 Summer 2026

Contest is running

4 weeks

Contestant



→ Submit?

Language: Java 21 64bit Choose file: Choose File No file chosen

Success!



Submit

```
3
1
```

input

Copy

```
8 2
1 7
7 3
3 6
6 5
5 2
2 8
8 4
8
6
4
2
1
7
5
8
3
```

output

Copy

```
4
1
8
1
2
5
2
3
```

input

Copy

```
1 1
1
1
```

output

Copy

```
1
```

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